# CPTG Pi-Branch Comparison Map
## Field Usefulness and Calibration-Stress Discipline

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**Abstract**

This report explains the purpose, current effectiveness, and field value of the CPTG geometric pi-branch comparison map. The map is useful because it allows a fixed CPTG-native branch to be translated into the observational coordinates used by modern cosmological surveys: supernova distance moduli, BAO distance-ruler ratios, CMB likelihood coordinates, BBN abundance summaries, and lensing/growth compression coordinates. The report also explains why calibration-stress layers must be kept out of the primary result unless they are part of the declared likelihood protocol from the start. The central principle is that observational nuisance parameters may be profiled or marginalized as part of the data pipeline, but locked CPTG branch parameters may not be moved to rescue a result. Each result must be labeled by the exact data layer tested: supernova distance shape, BAO distance-ruler coordinates, high-$\ell$ CMB profile scaffolding, or compressed support rows. Calibration-stress diagnostics remain scientifically useful, but they must be reported as diagnostics rather than allowed to redefine the primary validation claim.

## 1 Purpose of the Pi-Branch Comparison Map

The CPTG pi comparison map is not a separate theory and not a fitting layer. It is an observational interface. Its purpose is to take a fixed CPTG-native geometric branch and express that branch in the coordinates used by actual survey likelihoods and published data products [6, 7]. The map therefore separates three objects that are often mixed together in informal comparisons:

$$\begin{aligned}&\text{CPTG native geometric quantity}\\&\longrightarrow \text{comparison coordinate quantity}\\&\longrightarrow \text{reported observational summary.}\end{aligned}\tag{1}$$

The practical rule is the comparison-coordinate hierarchy used in the geometric pi paper [6]:

$$\begin{aligned}&\text{fix branch}\\&\longrightarrow \text{choose observational coordinate}\\&\longrightarrow \text{apply locked transport map}\\&\longrightarrow \text{compare without sector retuning.}\end{aligned}\tag{2}$$

This order is the reason the map is valuable. It prevents the comparison layer from becoming an invisible fit. If the branch is locked before the likelihood is evaluated, a successful comparison has real content. If the branch is adjusted sector by sector after looking at the data, the result is no longer a test of the pi branch.

## 2 Locked Branch Quantities

The central native and comparison quantities are fixed before any data are examined [6]. The current branch values used in the tests are:

$$p_C = \pi, \tag{3}$$

$$p_{\text{ac}} = 3 - \frac{\pi}{100} = 2.968584073464, \tag{4}$$

$$G_T = \frac{p_{\text{ac}}}{p_C} = 0.9449296585513721, \tag{5}$$

$$\sqrt{G_T} = 0.9720749243506758. \tag{6}$$

For the distance and expansion comparison layer, the pi branch uses:

$$B_\pi = 1 - \frac{1}{\pi}, \qquad T_\pi = \frac{1}{\pi}, \tag{7}$$

$$E_\pi(z)^2 = B_\pi + T_\pi(1+z)^\pi. \tag{8}$$

The raw geometric Hubble normalization and transported CMB comparison normalization are:

$$H_0^{(\pi)} = 69.4162507897\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{9}$$

$$H_{0,\text{CMB}}^{\text{CPTG}} = 67.4777967351\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{10}$$

The locked CMB row used in the Planck comparison is:

| Quantity | Locked CPTG comparison value |
|---|---|
| $H_0$ | $67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ |
| $\omega_b h^2$ | 0.022527857494 |
| $\omega_c h^2$ | 0.117685841620526 |
| $n_s$ | 0.968584073464 |
| $N_{\text{eff}}$ | 2.968584073464 |
| $A_s$ | $2.136283004441 \times 10^{-9}$ |
| $\tau$ | 0.058930875934 |
| $A_{\text{lens}}$ | 1 |

Table 1: Locked CPTG CMB acoustic-transport comparison row. These are not sector-by-sector fitted values.

The CMB gates behind this row are:

$$D_c = 0.9735353159758136, \tag{11}$$

$$A_{s,\mathrm{eff}} = A_{s,\mathrm{CPTG}} \frac{\pi}{3}, \tag{12}$$

$$\tau_{\mathrm{eff}} = \tau_{\mathrm{CPTG}} G_T^2, \tag{13}$$

$$A_{\mathrm{lens}} = 1. \tag{14}$$

These locked values define the theory object being tested. They are not allowed to move during an observational likelihood comparison unless a new branch is explicitly declared and tested as a different theory object.

## 3 Why This Map Is Useful in the Field

Modern observational cosmology does not usually compare theories in their native language. It compares data products in highly specific coordinates. Supernova data are compared as distance moduli. BAO data are compared through acoustic-ruler ratios such as $D_M/r_d$, $D_H/r_d$, or $D_V/r_d$. CMB data are compared through angular spectra and likelihood nuisance models. BBN is compared through abundance outputs and nuclear-network coordinates. Lensing and growth are often compressed into $S_8$, bandpower amplitudes, or response functions.

The pi comparison map is field-useful because it gives CPTG an audit-ready interface into each of these coordinate systems. A result can be written as a row:

$$\{\text{observable, CPTG map, data, covariance, residual, } \chi^2\text{, status}\}. \tag{15}$$

That structure makes the comparison falsifiable. It also makes it much easier for outside researchers to reproduce or challenge the result, because the branch values are locked before the data product is evaluated [8].

## 4 Scalable Pi-Based Comparison Versus Fitted Tuned Comparison

A pi-based scalable comparison is different from a fitted tuned comparison in both purpose and data handling. In the pi-based comparison, the CPTG branch is fixed before the observational data are evaluated. The branch values, transport factors, distance maps, acoustic gates, and comparison-coordinate rules are not adjusted separately for supernovae, BAO, CMB, BBN, or lensing. The same locked structure is carried into each survey coordinate system and then tested against the relevant data vector and covariance [6].

Operationally, the pi-based comparison begins with a fixed branch object,

$$\Theta_\pi = \{p_C,\, p_{\mathrm{ac}},\, G_T,\, D_c,\, A_s,\, T_\tau,\, H_0,\, r_d,\, N_{\mathrm{eff}},\, S_8\}_{\mathrm{locked}}\,. \tag{16}$$

The observational sector then determines the comparison coordinate. For example, supernovae use distance moduli, BAO uses acoustic ratios, and Planck uses CMB likelihood coordinates. The data-processing step is therefore a coordinate projection,

$$\Theta_\pi \longrightarrow m_s(\Theta_\pi, \nu_s), \tag{17}$$

where $m_s$ is the model vector in sector $s$, and $\nu_s$ denotes observational nuisance or control parameters belonging to that survey pipeline. These nuisance parameters describe calibration, foregrounds, intercepts, beam effects, or other measurement-system quantities. They are not new CPTG branch parameters.

The comparison statistic is then evaluated directly against the observed data vector:

$$\chi_s^2 = [d_s - m_s(\Theta_\pi, \nu_s)]^T C_s^{-1} [d_s - m_s(\Theta_\pi, \nu_s)]\,. \tag{18}$$

A fitted tuned comparison works differently. In that case, the theory-side parameters are allowed to move in order to improve the result for a particular dataset:

$$\Theta_s^* = \arg\min_{\Theta_s} \chi_s^2. \tag{19}$$

That procedure may be useful for phenomenological exploration, but it is a weaker test of branch coherence. If the parameter set changes from supernovae to BAO to CMB to lensing, the result no longer demonstrates that one geometric branch is being carried across sectors. It demonstrates only that each sector can be given its own optimized description.

The distinction is summarized in Table 2.

| Feature | Pi-based scalable comparison | Fitted tuned comparison |
|---|---|---|
| Theory branch | Fixed before data evaluation | Adjusted to improve a dataset-specific fit |
| Data processing | Data are expressed in their survey comparison coordinates | Theory parameters may be reselected for each dataset |
| Allowed movable quantities | Observational nuisance or control parameters only | Theory-side parameters may move unless externally constrained |
| Cross-sector meaning | Tests whether one branch survives multiple independent projections | Tests whether each sector can be separately accommodated |
| Falsifiability | Stronger, because the branch can fail without retuning | Weaker, because tensions can be absorbed by refitting |
| Field use | Suitable for audit tables, covariance tests, and survey-pipeline comparisons | Useful for exploratory modeling, but less decisive as a unified branch test |

Table 2: Difference between a scalable pi-based comparison and a fitted tuned comparison.

The advantage of the pi-based method is that it makes the comparison reproducible and falsifiable. Each sector receives the same locked branch, translated only into the observational language required by that sector. A failure in one sector cannot be hidden by changing the branch locally. A success across several sectors therefore has greater diagnostic value, because the same geometric structure has survived multiple independent data-processing environments.

This is also why observational nuisance profiling must be kept conceptually separate from theory retuning. Profiling supernova intercepts, BAO covariance coordinates, or Planck foreground and calibration nuisance terms improves the fairness of the measurement comparison. It allows each survey likelihood to be evaluated in its intended observational form. It does not alter the locked CPTG branch. By contrast, moving $p_{\rm ac}$, $G_T$, $D_c$, $T_\tau$, $A_s$, $N_{\rm eff}$, or $A_{\rm lens}$ to rescue a particular sector would convert the test into a fitted tuned comparison and would no longer test the original pi branch.

## 5 Current Real-Data Performance

The current tests show that the pi comparison map is not merely a cosmetic translation. It has produced real-data results across three major observational sectors: Pantheon+ supernova full covariance, DESI DR2 BAO covariance, and Planck 2018 high-$\ell$ Plik TTTEEE likelihood profiling [2, 3, 4, 1, 5].

### 5.1 Pantheon+ Full-Covariance Supernova Distance Shape

The supernova comparison uses the distance-coordinate layer from the geometric pi comparison map [6]:

$$D_M(z) = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')}, \tag{20}$$

$$D_L(z) = (1+z)D_M(z), \tag{21}$$

$$\mu(z) = 5\log_{10}\left(\frac{D_L(z)}{\mathrm{Mpc}}\right) + 25. \tag{22}$$

The Pantheon+SH0ES run used 1624 non-calibrator supernova rows after excluding 77 calibrator rows [2, 3]. The full statistical-plus-systematic covariance was used, and one additive magnitude/intercept nuisance was analytically marginalized. That means the test is a distance-shape comparison, not an absolute $H_0$ calibration claim.

| Model | $\chi^2$ | $\chi^2$/dof | $\Delta\chi^2$ vs Planck | Status |
|---|---|---|---|---|
| Best-shape LCDM control | 1456.974254 | 0.897704 | −1.040398 | control |
| Planck LCDM control | 1458.014652 | 0.898345 | 0.000000 | control |
| CPTG raw $\pi$ branch | 1458.058773 | 0.898373 | +0.044121 | pass |
| CPTG acoustic/clock branch | 1458.644867 | 0.898734 | +0.630215 | pass |
| DESI-like LCDM control | 1461.659069 | 0.900591 | +3.644417 | control row |

Table 3: Pantheon+SH0ES full-covariance supernova distance-shape performance.

The raw pi branch differs from the Planck LCDM control by only:

$$\Delta\chi^2_{\mathrm{raw}\,\pi-\mathrm{Planck}} = +0.044121. \tag{23}$$

As a likelihood ratio, this is:

$$\frac{\mathcal{L}_{\text{raw}\,\pi}}{\mathcal{L}_{\text{Planck}}} = \exp\left(-\frac{0.044121}{2}\right) = 0.97818. \tag{24}$$

This means the raw pi distance branch is about 97.8% as likely as the Planck LCDM control in this full-covariance distance-shape comparison. That is a strong sector result, but it remains a distance-shape result only.

### 5.2 DESI DR2 BAO Real-Data Vector and Covariance

The BAO comparison uses acoustic-ruler coordinates [6, 4]:

$$\frac{D_M(z)}{r_d}, \qquad \frac{D_H(z)}{r_d}, \qquad \frac{D_V(z)}{r_d}. \tag{25}$$

The DESI DR2 run used the 13-point BAO consensus vector and total covariance [4]. The raw pi controls failed, but the locked active acoustic branch passed without a fitted scale nuisance.

| Model | dof | $\chi^2$ | $\chi^2$/dof | Max diagonal pull |
|---|---|---|---|---|
| Raw pi native control | 13 | 537.728712 | 41.363747 | 10.395 |
| Raw pi transported control | 13 | 220.776424 | 16.982802 | 8.240 |
| Locked active acoustic branch | 13 | 10.639085 | 0.818391 | 1.626 |
| Active acoustic plus one scale diagnostic | 12 | 10.639085 | 0.886590 | 1.626 |

Table 4: DESI DR2 BAO real-data covariance performance. The one-scale row is diagnostic only; the locked active acoustic branch already passes without it.

The important lesson from DESI is that the comparison map selects the correct observational branch. The raw pi distance controls are not adequate for the BAO ruler vector, while the locked active-acoustic comparison branch is.

### 5.3 Planck 2018 High-$\ell$ Plik TTTEEE Likelihood

The Planck test is the most sensitive to methodology because Planck high-$\ell$ likelihoods contain many observational nuisance parameters [1, 5]. These include foreground amplitudes, calibration-like terms, beam corrections, point-source terms, and other pipeline quantities. Those parameters are not CPTG branch parameters. They belong to the observational likelihood machinery.

The pi-aligned Planck test therefore follows the comparison-coordinate rule in Ref. [6]: the CPTG cosmological row is locked, the Planck nuisance block is profiled under the same protocol used for the Planck control, and the comparison is made through the objective:

$$\text{objective} = -2\ln\mathcal{L}_{\text{Plik}} + \chi^2_{\text{nuisance priors}}. \tag{26}$$

The primary result from the pi-aligned single protocol is:

| Row | $-2\ln\mathcal{L}$ | Prior $\chi^2$ | Objective | $\Delta$ objective |
|---|---|---|---|---|
| Planck CAMB control | 2345.038424 | 1.033738 | 2346.072163 | 0.000000 |
| CPTG locked CMB row | 2347.939289 | 1.033738 | 2348.973028 | +2.900865 |

Table 5: Planck 2018 high-$\ell$ Plik TTTEEE pi-aligned profile scaffold.

The control-level criterion from the comparison map is:

$$\Delta\chi^2 \leq 5. \tag{27}$$

The decisive-stress criterion is:

$$\Delta\chi^2 > 10. \tag{28}$$

The Planck high-$\ell$ result therefore sits inside the control-level gate:

$$\Delta\chi^2_{\text{objective}} = +2.900865. \tag{29}$$

This should be described as a real high-$\ell$ Plik TTTEEE profile scaffold. It is not yet a full 47-dimensional posterior or a combined Planck high-$\ell$ plus lowE plus lensing validation.

## 6 Publication-Scope Claim Discipline

The present report is publishable only if each validation claim is tied to the exact data layer and likelihood protocol that produced it. A sector pass should not be promoted into an all-sector cosmology validation. The comparison map is strongest when every result is labeled by its implemented data product, covariance treatment, nuisance handling, and remaining limitation [6].

The publication-scope status should be read as follows:

| Sector | Implemented test | Proper claim | Not claimed |
|---|---|---|---|
| Pantheon+ | Full statistical-plus-systematic covariance distance-modulus comparison with one intercept nuisance | Full-covariance supernova distance-shape pass | Absolute $H_0$ calibration or full cosmology validation by itself |
| DESI DR2 BAO | Real BAO consensus vector and total covariance in acoustic-ruler coordinates | Distance-ruler coordinate pass for the locked active acoustic branch | DESI full-shape AP/RSD/growth validation |
| Planck TTTEEE | Real high-$\ell$ Plik TTTEEE likelihood with a pi-aligned nuisance-profile scaffold | Control-level high-$\ell$ CMB profile scaffold | Full 47-dimensional posterior, lowE, lensing, or combined-stack validation |
| BBN, $S_8$, and lensing summaries | Compressed or network-layer comparison rows from the comparison map | Supportive comparison-coordinate checks | Universal raw-likelihood closure across all growth and lensing data products |

Table 6: Publication-scope claim discipline for the current pi-comparison field report.

This table is important because it keeps successful layers from being overextended. Pantheon+ establishes a supernova distance-shape result. DESI DR2 BAO establishes a distance-ruler coordinate result. Planck

TTTEEE establishes a high-$\ell$ profile-scaffold result under the declared nuisance protocol. Compressed BBN, $S_8$, and lensing rows are useful support layers, but they do not replace full likelihood tests in their own sectors.

The same discipline applies to failures and stresses. A stress diagnostic should be reported, but it should not be allowed to redefine the primary result unless the diagnostic constraint is promoted into the declared likelihood protocol and the control and CPTG rows are both rerun consistently under that protocol.

## 7 Why Observational Nuisance Profiling Is Necessary

Fixed-nuisance comparisons are not the primary test for Planck high-$\ell$ likelihoods [1, 5]. A fixed nuisance vector is tuned to a particular control spectrum and observational calibration state. If the cosmological spectrum changes, the foreground and calibration nuisance model must be allowed to re-equilibrate. That is not theory retuning. It is how the observational likelihood is designed to be used.

The key distinction is:

$$\text{allowed to move : observational nuisance and control parameters,} \tag{30}$$

$$\text{not allowed to move : locked CPTG branch or comparison map parameters.} \tag{31}$$

This distinction improves measurement accuracy without weakening the theory test. It allows Planck foregrounds, calibration factors, and instrumental terms to play the same role for CPTG that they play for LCDM control spectra. But it does not allow $p_C$, $p_{\rm ac}$, $G_T$, $D_c$, $A_s = \pi/3$, $T_\tau = G_T^2$, $n_s$, $N_{\rm eff}$, or $A_{\rm lens} = 1$ to move.

In short:

$$\text{nuisance profiling} \neq \text{CPTG retuning.} \tag{32}$$

## 8 What a Calibration-Stress Layer Is

A calibration-stress layer is an additional diagnostic test that asks whether the profiled solution is sensitive to a calibration prior, calibration convention, or calibration-weighted variant of the likelihood protocol. It is not the same object as the primary likelihood comparison unless that calibration constraint is declared as part of the primary protocol from the beginning and applied symmetrically to both the control row and the CPTG row.

In the single pi-aligned Planck TTTEEE profile package, the primary result was:

$$\Delta\chi^2_{\rm primary} = +2.900865. \tag{33}$$

Optional calibration-stress diagnostics can still be useful because they identify where the profiled solution is sensitive to absolute calibration assumptions. However, a calibration-stress number should not be mixed into the primary result unless the full likelihood objective is redefined in advance and the nuisance block is reprofiled under that same objective for both the control spectrum and the locked CPTG spectrum.

## 9 Why Calibration Stress Is Not Primary

The importance of excluding calibration-stress layers from the primary result is not to hide tension. It is to keep the primary statistical object well-defined. A primary likelihood comparison must answer one clear question:

$$\text{How does the locked CPTG row perform under the declared likelihood protocol?} \tag{34}$$

A calibration-stress diagnostic asks a different question:

$$\text{How sensitive is the profiled solution to a calibration stress constraint?} \tag{35}$$

Those are both useful questions, but they are not the same question. Combining them without changing the label creates a confused result.

### 9.1 Reason 1: It Changes the Comparison Object

The primary Planck TTTEEE result used the official uploaded Plik likelihood code and data with a declared nuisance-profile protocol. Adding a separate calibration constraint after the fact changes the objective function. It becomes a different statistical object:

$$\text{primary objective} = -2\ln\mathcal{L}_{\text{Plik}} + \chi^2_{\text{nuisance}}, \tag{36}$$

while the calibration-stress objective is:

$$\text{stress objective} = -2\ln\mathcal{L}_{\text{Plik}} + \chi^2_{\text{nuisance}} + \chi^2_{A_{\text{Planck}}}. \tag{37}$$

A different objective function deserves a different label.

### 9.2 Reason 2: After-the-Fact Penalties Are Not Full Reprofiling

A calibration prior imposed after the profile does not let the full nuisance block re-equilibrate under that prior. It can overstate or misstate the stress because it evaluates a profiled point optimized under one protocol using a different objective.

If a calibration constraint is to be a primary likelihood ingredient, it must be included from the beginning:

$$\begin{aligned}
&\text{include calibration constraint from start}\\
&\longrightarrow \text{profile all nuisance consistently}\\
&\longrightarrow \text{compare control and CPTG with the same objective.}
\end{aligned} \tag{38}$$

Until that is done in a full declared protocol, the calibration-stress run should remain a diagnostic.

### 9.3 Reason 3: It Can Double-Count or Misplace Pipeline Assumptions

Planck high-$\ell$ likelihoods already contain calibration and foreground structure through their nuisance machinery. Adding a calibration-stress layer without declaring exactly how it is represented in the active likelihood can double-count a constraint or move the comparison into a different statistical protocol. That does not make the stress diagnostic illegitimate. It means it must be labeled as a diagnostic unless it is declared as part of the primary likelihood definition from the start.

### 9.4 Reason 4: It Would Let a Diagnostic Override the Main Result

A stress diagnostic is designed to identify sensitivity. It should not replace the primary result unless it is promoted into the primary protocol and rerun consistently. Otherwise, a narrow calibration sensitivity can mask the broader result: the locked CPTG row achieved a control-level high-$\ell$ TTTEEE profile result with $\Delta\chi^2 = +2.900865$.

### 9.5 Reason 5: Claim Discipline Protects Falsifiability

The comparison map is valuable because each claim means exactly what the implemented test means. Pantheon+ full covariance means a supernova distance-shape pass. DESI DR2 BAO covariance means a BAO ruler-coordinate pass. Planck high-$\ell$ Plik profile means a high-$\ell$ CMB profile scaffold. Calibration-stress means a calibration sensitivity diagnostic.

If these layers are merged into one ambiguous number, the result becomes harder to reproduce, harder to falsify, and easier to misrepresent.

## 10 Recommended Status Language

The most accurate status language is:

$$\text{Pantheon+ : full covariance distance shape pass.} \tag{39}$$

$$\text{DESI DR2 BAO : real data vector and covariance pass for the locked active acoustic branch.} \tag{40}$$

$$\text{Planck TTTEEE : control level high-}\ell\text{ Plik profile scaffold.} \tag{41}$$

$$\text{Calibration stress : optional diagnostic, not primary unless included from the start.} \tag{42}$$

The wording should avoid overclaiming. The Planck result should not be described as a final full-CMB validation until a full nuisance posterior or official pipeline-quality profile/posterior run is complete. But it is also wrong to demote the pi-aligned primary result because of an after-the-fact calibration-stress layer.

## 11 Field Value of the Pi-Based Comparison

The pi comparison map would be useful in the field for five reasons.

First, it creates a reproducible audit protocol. Every test begins from the same locked branch values and reports a transparent residual or likelihood shift.

Second, it allows direct comparison with standard survey products. The map speaks in the same coordinates used by Pantheon+, DESI BAO, Planck likelihoods, BBN summaries, and weak-lensing analyses.

Third, it prevents hidden fitting. The branch is not allowed to move from sector to sector. This is essential if CPTG is to be treated as a falsifiable framework rather than a post-hoc reconstruction.

Fourth, it separates validation layers. A supernova distance-shape pass is not automatically a CMB pass; a BAO pass is not automatically a full-shape AP/RSD pass; a Planck high-$\ell$ profile pass is not automatically a full lowE+lensing posterior pass. That separation makes the claim structure stronger, not weaker.

Fifth, it identifies where future work should focus. The current results suggest that distance and ruler sectors are strong, while Planck calibration and full nuisance-posterior behavior deserve deeper testing.

## 12 Limits and Next Steps

The pi comparison map is already effective as a comparison-coordinate framework, but field-grade acceptance would require independent reproduction. The highest-priority next steps are:

1. Run a full Planck TTTEEE nuisance posterior or high-quality profile with official nuisance priors included from the start [1, 5].
2. Extend to Planck lowE and lensing under a declared combined-stack protocol.
3. Repeat Pantheon+ with independent code and verify covariance alignment, calibrator removal, and intercept marginalization [2, 3].
4. Repeat DESI DR2 BAO using the official likelihood wrapper or an independent covariance implementation [4].
5. Keep DESI full-shape AP/growth labeled exploratory until official nuisance-preserving full-shape machinery is used.
6. Publish primary results and stress diagnostics as separate tables, never as a single mixed claim.

## 13 Conclusion

The CPTG pi-based comparison is useful because it converts a native geometric branch into observationally testable coordinates without allowing the theory to drift. The current real-data tests show strong but layer-specific performance: Pantheon+ full-covariance distance shape is essentially tied to the Planck LCDM control, DESI DR2 BAO passes under the locked active acoustic branch, and Planck high-$\ell$ Plik TTTEEE gives a control-level profile scaffold. These results should be reported as field-facing comparison-coordinate passes, not as a final all-sector cosmology validation.

The calibration-stress issue does not invalidate the primary Planck result. It demonstrates why comparison layers must be labeled carefully. A calibration-stress layer is valuable because it reveals sensitivity to calibration assumptions. But it is not the primary result unless it is part of the declared likelihood protocol from the beginning and applied symmetrically to both the CPTG and control rows.

The disciplined conclusion is therefore:

$$\text{The pi comparison map is an effective, falsifiable, field usable audit layer.} \tag{43}$$

$$\text{Calibration stress diagnostics should be reported, but not allowed to redefine the primary result.} \tag{44}$$

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